Started on	Saturday, 16 May 2026, 11:43 AM
State	Finished
Completed on	Saturday, 16 May 2026, 11:55 AM
Time taken	11 mins 14 secs
Marks	5.00/5.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1
Correct
Mark 1.00 out of 1.00

Based on the feature hierarchy of deep convolutional neural networks discussed in the presentation, what specific features do the **earlier layers** versus the **middle layers** typically extract, and how are they utilized during transfer learning?

- a. D) Earlier layers extract low-frequency noise, while middle layers extract semantic content; only earlier layers are frozen.
- b. C) Earlier layers extract shapes, while middle layers extract complex textures; both must be completely replaced for any new target dataset.
- c. A) Earlier layers extract task-specific classes, while middle layers extract general edges; only middle layers are frozen.
- d. B) Earlier layers extract basic edges, while middle layers extract shapes; both sets of layers are typically kept to leverage pre-learned general features while retraining task-specific layers. ✓

Your answer is correct.

The correct answer is:
B) Earlier layers extract basic edges, while middle layers extract shapes; both sets of layers are typically kept to leverage pre-learned general features while retraining task-specific layers.

Question 2
Correct
Mark 1.00 out of 1.00

According to the lecture text, what is an absolute architectural constraint regarding the input configuration when transferring a network from $\text{Task } 1$ to $\text{Task } 2$?

- a. C) Task 1 and Task 2 must share the same input setting configuration, such as spatial dimensions and channel counts. ✓
- b. D) Task 2 must use a completely different input preprocessing pipeline than Task 1.
- c. A) Task 1 and Task 2 must have the exact same number of target classification categories.
- d. B) Task 1 and Task 2 must possess an identical number of training samples.

Your answer is correct.

The correct answer is:
C) Task 1 and Task 2 must share the same input setting configuration, such as spatial dimensions and channel counts.

Question 3

Correct

Mark 1.00 out of 1.00

If you are building an image classifier with an extremely limited dataset and face strict resource limits, such as needing to complete training within a few minutes on a standard CPU, which strategy provides the optimal path forward according to the course guidelines?

- a. C) Fine-tuning the entire model (all layers) simultaneously with a high learning rate.
- b. B) Utilizing a pre-trained model with known weights, freezing the lower layers, and training only the final task-specific layers. ✓
- c. D) Replacing the early convolutional layers while freezing the upper fully connected layers completely.
- d. A) Reinitializing all network weights randomly and training the deep architecture from scratch.

Your answer is correct.

The correct answer is:
B) Utilizing a pre-trained model with known weights, freezing the lower layers, and training only the final task-specific layers.

Question 4

Correct

Mark 1.00 out of 1.00

The material identifies three distinct approaches to fine-tuning a pre-trained model. Suppose a machine learning engineer transitions from training only final layers to fine-tuning top layers and finally to fine-tuning the entire model. What are the primary trade-offs encountered along this spectrum?

- a. D) The model becomes less sensitive to input settings, reducing the need for identical image resolutions.
- b. B) Training speed increases, accuracy decreases, and the reliance on pre-trained weights increases.
- c. A) Training speed decreases, accuracy generally increases, and the requirement for a larger target dataset volume increases. ✓
- d. C) Computational resource requirements decrease, while the risk of underfitting increases.

Your answer is correct.

The correct answer is:
A) Training speed decreases, accuracy generally increases, and the requirement for a larger target dataset volume increases.

Question 5

Correct

Mark 1.00 out of 1.00

During transfer learning, if upper layers are modified by tweaking, dropping, or replacing them, what is the ultimate functional goal of this operation?

- a. B) To make the network completely linear and remove non-linear activations.
- b. D) To reduce the input dimensions so that the model can run without a CPU.
- c. C) To adapt the model's final representation mapping to the specific category label space of the new target task. ✓
- d. A) To alter the basic edge detection capabilities of the first layer.

Your answer is correct.

The correct answer is:
C) To adapt the model's final representation mapping to the specific category label space of the new target task.